

StealthShield

Zero-knowledge stealth addresses for everyday private finance on Solana.

Zcash Track · Public Goods Prize · Crypto World's Fair

THE THESIS

Every chain is public-by-default. Zcash proved the demand for privacy; StealthShield delivers the Zcash thesis cross-chain with a **working Solana reference implementation** built from real X25519/ed25519 cryptography.

CORE ARCHITECTURE

Stealth Derivation (X25519 ECDH)

Sender derives one-time $PK_{stealth} = PK_{spend} + sha256(shared_secret || index) \cdot G$. Only the recipient's viewing key can invert the shared secret. No sender→recipient linkability.

1-Byte View-Tag Scanning

$view_tag = sha256(shared_secret || index)[0]$ prunes non-matches in ms with **zero false positives** — thousands of announcements scanned at one X25519 + one SHA-256 each.

Anchor Rust Registry

Verified on-chain at `programs/solana_stealth_shield: register_meta_key, send_stealth_payment` (CPI transfer_checked, works with SPL **and** Token-2022 via `token_interface`), `sweep_stealth_funds`.

Real Client Cryptography

`src/services/stealthCrypto.ts` uses audited `@noble/curves X25519+ed25519`. Verified round-trip: sender-derived address == recipient-recovered address, view tags match.

EXTRAS

- Token-2022 confidential transfer extensions + privacy-preserving token wrappers integrated.
- On-chain verification harness + local-validator quickstart (agave 4.2.2 CI).
- Snapshots/announcements PDA content-addressed — no central indexer required.

WHY WE WIN

- **It's real:** audited primitives, a verifiable on-chain registry, and a passing test suite — not a spec.
- **Zcash home thesis delivered for Solana:** the exact problem Zcash track judges care about, with a path to confidential-transfer parity.
- **Public good:** open protocol, docs, permissive angle — eligible for the \$5K Public Good prize.